A1 / DESIGN AND MODELLING

Use-case descriptions

Centralized Workshop & Laboratory Allocation Platform

Actor	Responsibility
Student	Select a workshop, create a booking, retain its key and edit or delete the booking.
Workshop administrator	Review workshop requests, filter by date, change status and remarks, edit and delete records.
System administrator	Maintain master and workshop passcodes through system settings.
Desktop email application	Handle a mailto draft opened by SYNCLAB. Sending is a separate user action.

System boundary

The application runs on a desktop and keeps its relational data in a local SQLite file. Student access starts at the welcome screen. There is no student account login. Workshop administrator and system settings access use separate passcodes.

Five main use cases

UC-01 creation; UC-02 key-based editing; UC-03 administrator review; UC-04 date filtering; UC-05 configuration and correspondence.

Implementation basis: StudentWorkshopApp.pro and its compiled sources. Prepared 07 October 2026. All demonstration identities and bookings are fictional.

UC-01

Create a workshop allocation

Field	Description
Primary actor	Student
Precondition	Student needs a current or future booking and at least one free slot.
Main flow	Welcome > Allocate now > select workshop > calendar > choose slot(s) > Next > complete identity, supervisor and project > Submit.
Validation / rules	Required fields and email are validated. Database checks key uniqueness and slot conflicts inside a write transaction.
Postcondition	A Pending booking and slot rows persist. The application displays a four-digit allocation key.
Alternative / error path	Invalid input, a busy slot or a database write error displays a message and prevents a partial save.

Use-case text describes the active implementation. Email delivery and remote multi-user access are outside the current system boundary.

UC-02

Find and change an existing allocation

Field	Description
Primary actor	Student
Precondition	Student retains the allocation key.
Main flow	Welcome > Edit allocation > enter key > provide student index if a legacy duplicate key needs disambiguation > change schedule or form > Update.
Validation / rules	Edit flow verifies the key and excludes the current booking from availability checks.
Postcondition	Updated student allocation returns to Pending for review. Key and creation timestamp remain unchanged.
Alternative / error path	No unique matching record stops the flow. Deleting requires confirmation and removes child slot rows.

Use-case text describes the active implementation. Email delivery and remote multi-user access are outside the current system boundary.

UC-03

Review and manage workshop requests

Field	Description
Primary actor	Workshop administrator
Precondition	Administrator selects a workshop and enters its passcode.
Main flow	Calendar > Administrator > authenticate > choose a record > set status or remarks > use Edit or Delete as required.
Validation / rules	Only Pending, Approve and Reject are accepted. Re-activating a rejected booking checks for a slot conflict.
Postcondition	Status or remarks persist. Rejected bookings release their slots.
Alternative / error path	Wrong passcode displays Access denied. Conflicting re-approval fails without changing the stored status.

Use-case text describes the active implementation. Email delivery and remote multi-user access are outside the current system boundary.

UC-04

Inspect allocations by date

Field	Description
Primary actor	Student / administrator
Precondition	Database is ready and a workshop is selected.
Main flow	Calendar > select a date > inspect status counts and unavailable slots. Admin dashboard > choose date or Show all.
Validation / rules	Queries filter by category and ISO date. Active availability excludes Reject records.
Postcondition	Student sees schedule occupancy. Administrator sees matching records and dates.
Alternative / error path	An empty admin date produces an empty record list; student calendar shows free slots and no counts. SQL failures are available through getLastError().

Use-case text describes the active implementation. Email delivery and remote multi-user access are outside the current system boundary.

UC-05

Configure access and compose correspondence

Field	Description
Primary actor	System administrator / workshop administrator
Precondition	Authorized administrator has the relevant passcode.
Main flow	Welcome > System settings > master passcode > change access settings. Admin dashboard > Settings or Pending/Reject email action.
Validation / rules	QSettings persists passcodes and email configuration. Status email actions open a mailto draft.
Postcondition	Configuration is stored locally. An installed email application can review and send the prepared draft.
Alternative / error path	Wrong passcode blocks access. A missing email handler prevents completion of the draft workflow.

Use-case text describes the active implementation. Email delivery and remote multi-user access are outside the current system boundary.